Glucose Monitoring System

1. Project Overview

The **Glucose Monitoring System** is a simulation-based embedded system designed to demonstrate the basic process of measuring and displaying blood glucose levels.

The system uses an **ATmega32 microcontroller** to receive an analog signal, convert it into a digital value using its built-in **10-bit ADC**, process the result, and display the corresponding glucose level on an LCD.

A **potentiometer** will be used in Proteus to simulate the variable analog output of a glucose sensor.

2. Project Objectives

Simulate a basic glucose monitoring system using an embedded microcontroller.

Interface an analog input with the **ATmega32 ADC**.

Convert the ADC reading into a corresponding glucose value.

Display the glucose level on an LCD.

Apply Embedded C programming and driver-based software architecture.

Practice hardware/software integration using **Proteus and Eclipse**.

3. Hardware & Software Components

Hardware

- **ATmega32 Microcontroller**
- **Potentiometer** – used to simulate the glucose sensor output
- **LCD** – used to display the glucose level

Software

- **Embedded C**
- **Eclipse** – for software development
- **Proteus** – for circuit design and simulation

4. System Working Principle

- The system will operate according to the following sequence:
- The potentiometer generates an analog voltage.
- The analog voltage is connected to an ADC input of the ATmega32.
- The ATmega32 starts an ADC conversion.
- The ADC converts the analog voltage into a 10-bit digital value.
- The application processes the ADC result.

- The digital value is mapped to a corresponding glucose level.
- The calculated glucose level is displayed on the LCD.
- The system continuously repeats the process to provide updated readings.

System Flow

Analog Input → ADC Conversion → Digital ADC Value(0–1023) → Glucose Calculation →Glucose Level

→ LCD Display

5. Expected Outcome

The final Proteus simulation is expected to demonstrate a functional glucose monitoring system in which changing the potentiometer value produces a corresponding change in the displayed glucose level.

Side Note : The project is intended for ***educational and simulation purposes only*** *and does not represent a real medical device or provide medical measurements.*